

Rupam Sah

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Professional Profile

Lecturer in Computer Science with an M.Tech in Computer Science and Engineering from NIT Jalandhar, Punjab, India, and practical experience in machine learning, deep learning, and computationally intensive data processing workloads. My dissertation focused on developing AI-driven approaches for automated pancreas segmentation from abdominal CT images, involving deep learning architectures, attention mechanisms, image preprocessing, and computational experimentation with large-scale medical imaging datasets. This work developed my interest in designing efficient and reliable computational methods for complex scientific and data-intensive problems. I am now particularly interested to explore in High Performance Computing, machine learning and deep learning, numerical methods, and efficient computational techniques, while retaining a strong background in medical image analysis and scientific programming.

Academic Appointment

Sai University, School of Computing and Data Science

Chennai, India

Lecturer (Computer Science)

July 2026 - Present

- Design and deliver undergraduate modules in **Web Technology** and **Design and Analysis of Algorithms (DAA)** through lectures, tutorials, laboratory sessions, coursework, and examinations.
- Develop lecture materials, laboratory exercises, programming assignments, assessments, and marking schemes covering web technologies, client-server programming, internet-based application development, algorithm design, complexity analysis, and fundamental problem-solving techniques.
- Conduct hands-on laboratory sessions for **Web Technology** and **DAA**, guiding students in implementing web applications, programming solutions, and algorithms while analysing their correctness, efficiency, and computational complexity.
- Provide academic advising and student support, assess student performance through coursework, laboratory work, and examinations, and deliver timely, constructive feedback to strengthen conceptual understanding and practical programming skills.
- Contribute to curriculum development, course planning, assessment activities, student mentoring, and academic administration within the School of Computing and Data Science.

Dayananda Sagar University, School of Engineering

Bengaluru, India

Assistant Professor (Computer Science)

July 2025 - June 2026

Publications

A Hybrid Framework for Automated Pancreas Segmentation in Abdominal CT Imaging | First author | PReMI 2025, IIT Delhi-Springer LNCS

- Co-authored with Prof. Dr. Renu Dhir; developed from my M.Tech thesis research on deep-learning-based pancreas segmentation from CT imaging.

AI-Driven Approaches for Improved Detection and Diagnosis of Pancreatic Cancer | First author | ICAIHC 2025, NIT Raipur - IEEE Xplore

- Co-authored with Prof. Dr. Renu Dhir; extended my segmentation work toward AI-assisted detection and diagnosis of pancreatic cancer.

Research Experience

M.Tech Thesis Project, NIT Jalandhar

Punjab, India

Supervised by Prof. Dr. Renu Dhir

2024 - 2025

- Proposed and developed **ESAT U-Net**, a deep learning framework engineered to enhance feature representation in complex biomedical pancreas segmentation.
- Strategically integrated Squeeze-and-Excitation Enhancement Blocks alongside Attention-Gated Enhancement Blocks to capture fine-grained spatial dependencies.
- Evaluated the framework on the benchmark NIH Pancreas CT Dataset, establishing a 4.78% increase in Dice Similarity Coefficient relative to baseline MAEU-Net architectures.

Projects

Student Result Analyzer | MongoDB, Express.js, React.js, Node.js (MERN Stack)

- Built an academic analytical pipeline using the MERN Stack for real-time tracking of student milestones alongside clean, automated report generation.
- Designed interactive, custom UI visualization dashboards to support institutional decision-making.

Certifications & Achievements

GATE (CSE) Qualified - 2023.

NPTEL Certification in Data Analytics with Python.

Udemy Certification: The Complete Web Development Bootcamp (Instructor: Dr. Angela Yu).

Academic Leadership & Responsibilities

Placement Representative (CSE), NIT Jalandhar, 2024 - 2025.

Teaching Assistant (TA), Department of CSE, NIT Jalandhar, 2023 - 2025.

Convenor, Social Service Club, Tezpur University Students' Council, 2021 - 2022.

Education

National Institute of Technology (NIT), Jalandhar <i>M.Tech in Computer Science and Engineering, CGPA: 7.66/10.0</i>	Punjab, India 2023 – 2025
Tezpur University (A Central University) <i>B.Tech in Computer Science and Engineering, CGPA: 7.48/10.0</i>	Assam, India 2019 – 2023

Technical Skills

Programming Languages: Python, C, C++, SQL
Core Domains: Data Structure and Algorithms, Machine Learning, Deep Learning
Tools & Frameworks: PyTorch, TensorFlow, Git, GitHub, Google Colab, Jupyter Notebook, VS Code
Operating Systems: Linux, Windows